

Education

Swarthmore College | GPA: 3.64/4.0

Expected May 2028

- Bachelor of Science in Engineering
- Bachelor of Arts in Astrophysics
- Relevant Coursework: Analytical Dynamics, Thermofluid Mechanics, Linear Physical Systems Analysis, Stochastic & Numerical Methods, Electricity & Magnetism, Thermodynamics, Differential Equations

Work Experience

Plasma Systems Engineering Researcher | Swarthmore College

May 2026 – Present

- Designed, fabricated, and integrated a LaB₆ cathode assembly for an experimental plasma device that increased plasma density by 100× relative to previous designs
- Performed finite element thermal analysis of plasma-facing components under 1 kW heating loads, optimizing heat transfer to maximize thermionic emission while maintaining material operating limits
- Demonstrated sustained cathode operation at 2000 K and validated performance with diagnostic probes following system integration.

Biomimetic Robotics Researcher | Swarthmore College

Oct 2024 – Feb 2026

- Developed a biomimetic swimming robot modeled after barnacle nauplius kinematics for operation in high-viscosity environments, integrating a biologically accurate carapace to study swimming efficiency and fluid mechanics
- Designed and implemented a light-activated motor control circuit with adjustable sensitivity using LTSpice to reliably allow contactless activation of the mechanism while submerged in viscous fluids
- Modeled a compact clockwork arm mechanism in Fusion 360, designed with an internal slider to eliminate piston effects and improve hydrodynamics, then successfully translated digital model into a functional physical system
- Analyzed pivot geometries to match scaled biological motion accurate to within 3% of true physiology and achieved a 20% reduction in overall robot height from previous models while preserving mechanical integrity and sealing capabilities

Physics Teaching Assistant | Swarthmore College

Fall 2025, Fall 2026

- Supported instruction for 80+ students in foundational courses for physics majors covering classical mechanics, optics, and quantum mechanics
- Led over 30 weekly problem-solving sessions and provided one-on-one tutoring to over 20 students

Projects

Reaction Wheel Attitude Control System

Mar 2026 – Present

- Designed and fabricated a reaction wheel testbed to demonstrate single-axis spacecraft attitude control and angular momentum management
- Applied Euler's equations, inertia tensor analysis, and conservation of angular momentum to develop MATLAB and Python simulations of rigid-body motion
- Validated experimental rotational behavior against analytical and numerical predictions, demonstrating agreement between theoretical spacecraft dynamics models and physical system performance

Skills

Programming: Python, MATLAB, Simulink, Java, SQL

CAD & Analysis: SolidWorks, Fusion 360, Onshape, FEA, Thermal Analysis